STROBE Statement—Checklist of Items for Observational, Functional Genomic, and Interventional Studies

Applying Causal Machine Learning and Multi-Omics Integration to Unravel Gene Regulatory Networks in Triple-Negative Breast Cancer (TNBC)

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Section / Topic	Item #	STROBE Checklist Recommendation	Detailed Implementation in TNBC Causal ML Manuscript	Manuscript Location
Title and Abstract	1a	Indicate study's design with a commonly used term in title/abstract	Title and Abstract explicitly indicate the study design: genome-scale single-cell CRISPRi perturbation screening, multi-omics epigenetic prior integration, causal DAG triangulation, and clinical survival validation in human TNBC cohorts.	Page 1 (Title, Abstract)
Title and Abstract	1b	Provide an informative and balanced summary of what was done and found	Structured abstract delineates clinical rationale (TNBC chemo-refractoriness); experimental scale (GSE224986; 95,678 cells, 11,975 sgRNAs, 819 super-enhancers); causal triangulation methodology; 6,366,350 reconstructed causal paths; prioritized master drivers (CTCF, POLR2A, EP300, JUND, MYC, SIN3A, SP1, FOS, HDAC2, MAX); enriched Hallmark pathways (EMT, E2F, G2M, Hypoxia); non-linear SHAP threshold switches; benchmarking vs GENIE3/co-expression (AUPRC = 0.875, AUROC = 0.900 vs 0.632 / 0.681); in silico driver KO validation (R^2 = 0.473, r = 0.688, p = 3.37×10^{-122}, N = 864); clinical survival in TCGA-BRCA (n = 160) and METABRIC (n = 320); and triple epigenetic combination therapy (BETi + HATi + HDACi) collapsing 93% of oncogenic circuitry.	Pages 1–2 (Abstract)
Introduction - Background	2	Explain scientific background and rationale for investigation	Sections 1.1–1.3: Aggressive clinical trajectory and lack of targeted receptor therapies in TNBC; reliance on dense distal super-enhancers; mathematical proof that observational transcriptomic GRNs (GENIE3, SCENIC) suffer high false discovery rates (>45% FDR) due to single-cell dropout zero-inflation; and necessity of interventional do-calculus and functional single-cell CRISPRi.	Pages 2–4 (Sections 1.1–1.3)
Introduction - Objectives	3	State specific objectives, including any prespecified hypotheses	Section 1.4: Defines 6 primary investigative objectives: (i) construct tripartite causal GRN (TF → Enhancer → Target Gene); (ii) prioritize master drivers via PageRank and MRS; (iii) decode non-linear cooperativity via Explainable AI (SHAP); (iv) benchmark against state-of-the-art GRN algorithms; (v) validate clinical prognosis in independent patient cohorts; and (vi) map synergistic epigenetic combination therapies.	Page 4 (Section 1.4)
Methods - Study Design	4	Present key elements of study design early in the paper	Sections 2.1–2.4: Comprehensive end-to-end causal machine learning workflow uniting physical chromatin occupancy (ENCODE Factorbook ChIP-seq) with randomized interventional perturbation responses (single-cell CRISPRi / dCas9-KRAB).	Pages 4–6 (Sections 2.1–2.4)
Methods - Setting	5	Describe setting, locations, relevant dates, exposure periods	Section 2.1: Preclinical cellular model MDA-MB-231 (human mesenchymal basal-like TNBC); genome-scale single-cell CRISPRi Perturb-seq screen across 8 independent 10x Genomics Chromium 3' sequencing lanes (GEO accession GSE224986).	Pages 4–5 (Section 2.1)

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Methods - Participants / Cells	6a	Give eligibility criteria, sources, and selection methods	Section 2.1: Stringent single-cell quality control thresholds: cells retaining >= 200 detected genes and <= 15% mitochondrial reads; genes expressed in >= 3 cells; resulting in 95,678 high-quality single cells and 32,799 reliably expressed gene features.	Pages 4–5, 8–10 (Section 2.1; Table 1; Figure 1)
Methods - Variables	7	Clearly define all outcomes, exposures, predictors, confounders	Sections 2.1–2.4: Exposure = targeted dCas9-KRAB single-guide enhancer silencing (819 somatic mutation loci); Outcome = log2 fold change and Welch t-statistic of downstream target genes; Predictors = ChIP peak confidence score C_ChIP(i, e) in [0, 1000] and causal weight W_causal; Potential Confounders = technical dropout zeros, 10x sequencing lane batches, and multi-guide epistatic crosstalk.	Pages 4–6 (Sections 2.1, 2.3, 2.4; Table 1)
Methods - Data Sources	8	Give sources of data and details of measurement methods	Sections 2.1, 2.2, 2.7, 5: GEO GSE224986 (Perturb-seq), GEO GSE223812 (enhancer coordinates), ENCODE Factorbook / UCSC encRegTfbsClustered (ChIP-seq priors across 340 TFs), TCGA-BRCA (n = 160 TNBC) and METABRIC (n = 320 TNBC) clinical cohorts, and DGIdb/ChEMBL/ClinicalTrials.gov for therapeutics.	Pages 4–5, 7–8, 21–22 (Sections 2.1, 2.2, 2.7, 5)
Methods - Bias Control	9	Describe any efforts to address potential sources of bias	Sections 2.1, 2.4, 4.1: Unconfounded causal identifiability enforced by evaluating single-guide perturbation cells (n = 3,930) strictly against unperturbed negative-control cells (n = 7,767), breaking multi-guide epistatic confounding; observational dropout zeros overcome by requiring physical ChIP-seq binding priors; lane batch effects controlled; acyclicity misspecification avoided by modeling multigraph topology rather than global continuous DAG inversion (Section 4.1).	Pages 4–6, 18–19 (Sections 2.1, 2.4, 4.1; Supplementary Note 2)
Methods - Study Size	10	Explain how study size was arrived at	Section 2.1: Complete experimental library of 95,678 single cells, 11,975 unique sgRNAs, 819 targeted super-enhancers, and 32,799 gene features; stratified into single-guide (n = 3,930), multi-guide (n = 83,981), and negative control (n = 7,767) cohorts.	Pages 4–5, 8–10 (Section 2.1; Table 1; Figure 1d)
Methods - Quantitative Variables	11	Explain how quantitative variables were handled in analyses	Sections 2.3–2.6: Library size normalization to 10,000 counts per cell (CP10k); log-transformation X_norm = ln(1 + CP10k); vectorized Welch t-statistic computation with Satterthwaite degrees of freedom v; composite causal weight scaling W_causal = C_ChIP × |log2 FC| × |t|.	Pages 5–6 (Sections 2.3, 2.4; Supplementary Notes 1, 3)
Methods - Statistical Methods	12a	Describe all statistical methods, including confounder control	Sections 2.3–2.7: Fast vectorized two-sample Welch t-test (Supplementary Note 3); Google PageRank graph centrality (damping factor $\alpha = 0.85$; Supplementary Note 4); Gradient Boosted Decision Trees (XGBoost Regressor; 100 estimators, depth = 4); exact TreeExplainer SHAP attribution (Supplementary Note 5); multivariable Cox proportional hazards modeling adjusted for age, tumor stage, and nodal status.	Pages 5–8 (Sections 2.3–2.7; Supplementary Notes 3–5)
Methods - Subgroups & Interactions	12b	Describe methods examining subgroups and interactions	Sections 2.3, 2.8: Unsupervised Leiden community detection (resolution = 0.5) resolving 12 distinct phenotypic states (Clusters 0 to 11); multi-factor combinatorial epigenetic perturbation simulations (BETi + HATi + HDACi).	Pages 5–6, 8, 17 (Sections 2.3, 2.8; Figures 1c, 8)
Methods - Missing Data Handling	12c	Explain how missing data were addressed	Sections 2.1, 2.4: Zero-inflation handled naturally via interventional differential testing without artificial expression imputation; chunked Compressed Sparse Row (CSR) matrix representation in h5py and AnnData.	Pages 4–6 (Sections 2.1, 2.4)

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Methods - Sensitivity & Cross-Validation	12d	Describe sensitivity analyses and cross-validation methods	Sections 2.6, 2.7: 5-fold cross-validation in XGBoost models (test $R^2 = 0.742$); empirical held-out CRISPRi ground-truth testing across multiple effect size thresholds (|log2 FC| >= 0.50, |t| >= 3.0); in silico driver knockout validation across N = 864 target genes.	Pages 7–8, 15–17 (Sections 2.6, 2.7, 3.7, 3.8; Table 6; Figure 7)
Results - Participants / Cells	13a	Report numbers of individuals/cells at each stage of study	Section 3.1: 95,678 single cells evaluated post-QC; 32,799 expressing genes; stratified across 12 Leiden clusters and 8 experimental sequencing lanes; 3,930 single-guide cells, 83,981 multi-guide cells, 7,767 negative-control cells.	Pages 8–10 (Section 3.1; Table 1; Figure 1)
Results - Descriptive Data	14a	Give characteristics of study participants/cells	Section 3.1, Table 1, Figure 1: Median 6,519 UMIs per cell (Figure 1a); median 2,138 detected genes per cell (Figure 1b); 819 targeted super-enhancer loci across 11,975 sgRNAs.	Pages 8–10 (Section 3.1; Table 1; Figure 1a–d)
Results - Outcome Data	15	Report numbers of outcome events or summary measures	Sections 3.2, 3.3: 15,059 physical TF–enhancer prior binding events across 161 unique TFs (Figure 2a; Table 2); 1,110,566 candidate interventional edges identified via vectorized Welch testing (Figure 3; Supplementary Data 2).	Pages 10–11 (Sections 3.2, 3.3; Table 2; Figures 2, 3)
Results - Main Findings	16a	Give unadjusted and adjusted effect estimates and precision	Sections 3.4–3.6: Reconstructed 6,366,350 tripartite causal paths (Supplementary Data 3); prioritized master drivers: CTCF (118.7k edges, MRS = 0.986), POLR2A (110.5k, MRS = 0.983), EP300 (67.3k, MRS = 0.809), JUND (75.3k, MRS = 0.806), MYC (60.9k, MRS = 0.768), SIN3A, SP1, FOS, HDAC2, MAX (Tables 3, 4; Figures 4, 5); non-linear TreeExplainer SHAP attribution ranking JUND (0.0296) and MYC (0.0174) with threshold dynamics (Table 5; Figure 6).	Pages 11–15 (Sections 3.4–3.6; Tables 3–5; Figures 4–6)
Results - Benchmarking & Validation	17	Report other analyses done (sensitivity, benchmarking, validation)	Sections 3.7–3.9: Triangulated Causal ML achieved AUROC = 0.8999, AUPRC = 0.8745, Top 25% Precision = 0.9013, FDR = 0.0987 vs ChIP-seq alone (AUROC 0.8890 / AUPRC 0.7228 / FDR 0.2440), Co-expression (0.7292 / 0.6814 / 0.3227), and GENIE3 (0.6698 / 0.6323 / 0.3573) (Table 6; Figure 7a,b); in silico driver KO validation yielded Pearson r = 0.688, $R^2 = 0.473$ ($p = 3.37 \times 10^{-122}$, N = 864 genes; Figure 7c); clinical Cox survival modeling in TCGA-BRCA (n = 160) and METABRIC (n = 320) confirmed significant risk for JUND (HR = 1.88, $p < 0.001$), MYC (HR = 1.74, $p < 0.001$), FOS (HR = 1.62, $p = 0.001$), EP300 (HR = 1.45, $p = 0.006$), and MAX (HR = 1.36, $p = 0.019$) (Figure 7d); in silico combinatorial therapy demonstrated 93% network collapse under BETi + HATi + HDACi vs 38% under chemotherapy (Figure 8; Table 7).	Pages 15–18 (Sections 3.7–3.9; Tables 6, 7; Figures 7, 8)
Discussion - Key Results & Conclusion	18	Summarize key results with reference to study objectives	Sections 4.1–4.4, Section 5 (Conclusion): Successfully triangulated 6.36M directed causal edges; resolved mathematical failure of observational GRNs (>45% FDR); established Pearlian interventional identifiability; unmasked non-linear AP-1/MYC switch dynamics; in silico knockout validation ($R^2 = 0.473$); and formulated tri-modal epigenetic combination therapy (BETi + HATi + HDACi) collapsing 93.4% of oncogenic super-enhancers while de-repressing tumor suppressors.	Pages 18–20, 21 (Sections 4.1–4.4; Section 5 Conclusion)
Discussion - Limitations	19	Discuss limitations of study (potential bias, imprecision)	Section 4.5: Reliance on preclinical in vitro cell line (MDA-MB-231) requiring future expansion into patient-derived organoids (PDOs); targeted silencing of non-coding enhancers rather than direct genetic knockouts of upstream TF coding genes; absence of single-cell Hi-C/Micro-C 3D looping data; computational rather than in vivo clinical trial testing of combination regimens.	Pages 20–21 (Section 4.5)

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Discussion - Interpretation	20	Give a cautious overall interpretation of results	Sections 4.2–4.4, Section 5 (Conclusion): Triangulating randomized Perturb-seq interventions with physical ChIP-seq priors provides a mathematically principled, biologically validated paradigm for discovering causal oncogenic drivers and designing rational epigenetic combination therapies.	Pages 19–21 (Sections 4.2–4.4; Section 5 Conclusion)
Discussion - Generalisability	21	Discuss the generalisability (external validity) of study results	Sections 4.4, 4.5: Demonstrates broad applicability of the causal triangulation framework across other treatment-refractory cancers characterized by super-enhancer addiction; clinical validation across independent patient cohorts (TCGA-BRCA, METABRIC) confirms prognostic generalisability beyond cell line models.	Pages 20–21 (Sections 4.4, 4.5)
Funding & Data/Code Availability	22	Give source of funding, role of funders, and accessibility of data/code	Section 5 (Data and Code Availability) & Declarations: Supported by National Science & Technology (NST) Fellowship (2025–2026), Ministry of Science and Technology, Government of the People's Republic of Bangladesh (Awardee: Omar Faroque). All code, notebooks, and models are available under MIT License at https://github.com/Omarrfaroqueiu79/tnbc-causal-grn; Zenodo DOI: 10.5281/zenodo.10892345; GEO accessions GSE224986 and GSE223812.	Pages 21–22 (Section 5 Data and Code Availability; Declarations)

Note: This checklist adheres to the STROBE (Strengthening the Reporting of Observational Studies in Epidemiology) guidelines adapted for molecular epidemiology, high-throughput single-cell functional genomics, and interventional causal discovery. All metrics, cohort dimensions, and experimental parameters strictly match the 29-page primary manuscript text, figures, tables, Section 5 (Conclusion), and Section 5 (Data and Code Availability).